

Chemotherapy-based control of schistosomiasis haematobia. II. Metrifonate vs. praziquantel in control of infection-associated morbidity.

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To determine the relative efficacy of metrifonate and praziquantel in controlling urinary tract morbidity due to *Schistosoma haematobium* infection, a random allocation treatment trial was performed among 1,813 school age *S. haematobium*-infected children from the Msambweni area of Coast Province, Kenya. Following baseline examination for infection, hematuria, proteinuria, and ultrasonographic urinary tract abnormalities, oral treatment with either metrifonate (10 mg/kg, repeated at 4 month intervals) or praziquantel (1 dose of 40 mg/kg) was given to infected subjects. Prevalence of morbidity was reassessed 12 months later for each treatment group. Results indicated equivalent patient improvement in response to either regimen: prevalence of hematuria fell from 75% to 17% after either praziquantel or metrifonate therapy. Similarly, prevalence of proteinuria was significantly reduced from 73% to 29% (metrifonate) or 27% (praziquantel) after therapy. Metrifonate and praziquantel caused similar reductions in bladder granulomata and bladder thickening; however, no reduction in hydronephrosis was noted with either drug. Analysis of outcomes in population subgroups defined by age, sex, pretreatment intensity of infection, or severity of pretreatment morbidity showed no consistent advantage for either drug. In this endemic area, both agents provide effective control of morbidity due to urinary schistosomiasis.

Schistosomiasis in an American medical investigator.

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A 42-year-old American male researcher contracted schistosomiasis from environmental sources in the course of his observations on human behavior in Upper Egypt. After a long asymptomatic period, he developed various symptoms and *Schistosoma haematobium* was found in a urine examination. After treatment with Metrifonate, urine examination became negative. However, abdominal pain persisted and most diagnostic tests were negative. Colonoscopic examination and biopsy of the mucosa revealed schistosomiasis. Treatment with Praziquantel was thoroughly effective in clearing the persistent *Schistosoma haematobium* infection. It is necessary to maintain a high index of suspicion in cases of potential schistosomiasis. The availability of nontoxic treatment is discussed.

Computer-Aided Discovery of *Abrus precatorius* Compounds With Anti-Schistosomal Potential.

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Schistosomiasis, which causes over 200 000 deaths annually, has since the 1970s been controlled by praziquantel. The reliance on a single drug to combat schistosomiasis, and reports of laboratory resistance to the drug, has created an urgent need in the scientific community to develop new chemotherapies to complement or supplement praziquantel. Medicinal plants are a potential reservoir of compounds with schistosomicidal activity. In the current study, we carried out computer-aided screening of *Abrus precatorius* compounds to discover compounds with potential to inhibit *Schistosoma mansoni* purine nucleoside phosphorylase (SmPNP). Thus, 99 compounds retrieved from Lotus Natural Compounds Database were docked into the active site of SmPNP. The top-ranked compounds were subjected to

Lipinski's druglikeness and toxicity risk predictions. Three lead compounds, abrusogenin, cirsimaritin and hispidulin, were identified as having high binding affinities, favourable interactions with SmPNP active site residues and good toxicity risk prediction results. Molecular dynamics (MD) simulations were used to assess the stability of the interactions of these lead compounds with SmPNP. Collectively, analyses of the MD trajectories confirms that the lead compounds bound and interacted stably with active site residues of SmPNP. We conclude that abrusogenin, cirsimaritin and hispidulin could serve as hit compounds for the development of new antischistosomal drugs, based on plant-derived natural products. However, experimental studies are required to further evaluate the potentials of these compounds as possible therapeutics against schistosomiasis.

Efficacy of Praziquantel in Treating *Schistosoma haematobium* Infection Among Ethiopian Children.

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Background/Objectives: Praziquantel is a cornerstone of schistosomiasis control and elimination efforts. Continued surveillance of praziquantel efficacy is needed to monitor for the development of resistance, as well as to help public health officials gauge the effect of mass praziquantel administration on schistosomiasis control in communities, since it is the only drug used in schistosomiasis control programs. The objective of this study was to assess the praziquantel cure rate and egg reduction rate against urogenital schistosomiasis. **Methods:** This study enrolled 977 children from 12 villages in Afar and Gambella, Ethiopia, who provided urine samples that were checked for *Schistosoma haematobium* infection at baseline using urine filtration microscopy. Infected individuals were provided a single dose of praziquantel (40 mg/kg body weight) and retested six weeks post-treatment. **Results:** *S. haematobium* was recovered from baseline urine specimens in 177 of 977 (18%) participating children. One hundred six of these children completed therapy and presented for subsequent evaluation at six weeks; 91 children were egg-free. The egg reduction rate was 97%; changes in egg burden among the 15 children who did not achieve cure varied widely. Cure rates were better among children with light-intensity infections. No significant differences in egg reduction rates were found based on the demographic variables examined. **Conclusions:** Standard praziquantel monotherapy remains an effective treatment against urogenital schistosomiasis in Ethiopia.

Drug metabolism and pharmacokinetics of praziquantel: A review of variable drug exposure during schistosomiasis treatment in human hosts and experimental models.

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Schistosomiasis control is heavily reliant on the drug praziquantel (PZQ), which is used as preventive chemotherapy as part of national helminth control strategies. Given the heavy reliance on PZQ for mass drug administration, there has been considerable research on the potential of parasites developing resistance to the drug, resulting in decreased drug efficacy. However, there have been comparatively fewer studies of other factors that can potentially alter PZQ efficacy. Here, we investigate whether host PZQ metabolism contributes towards variable cure rates. We evaluate factors that can influence the metabolism of PZQ and the resultant effect on the efficacy of PZQ treatment to determine factors that potentially influence an individual's response to the drug. The literature search was directed at published

studies from three online databases: Web of Science, PubMed, and EMBASE. The search terms for the review comprised of ([praziquantel OR PZQ] AND [schistosom* OR bilharzia] AND [pharmaco*]) and included studies evaluating PZQ metabolism. Publications were categorised into pharmacokinetics, drug-drug interactions, pharmacogenetics, and metabolite analysis. Forty publications describing human and experimental studies fitted the inclusion criteria and were subjected to data extraction and analysis. The analyses showed that variable exposure to PZQ was associated with alterations in the liver's capacity to metabolise PZQ and observed drug-drug interactions. Other factors influencing the efficacy of PZQ were brand, formulation, and co-administered food. Although some work has been performed on metabolite identification, there was minimal information on PZQ's metabolic pathway, and no pharmacogenetics studies were identified. The study indicated that in both human and experimental studies alterations in the liver's capacity to metabolise PZQ as well as drug-drug interactions affected systemic levels of PZQ that could result in variable cure rates. The study confirmed previous findings of higher antischistosomal activity of (R)-PZQ enantiomer when administered alone compared to the racemate at the same dose as well as improved efficacy when the drug is administered with food. The study also highlighted the need for more comprehensive studies of the PZQ metabolic pathway and PZQ pharmacogenetic studies in humans.

[Clinical observation on the treatment of 1,627 cases of schistosomiasis haematobia with praziquantel of different dosages].

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A total of 1,627 cases of schistosomiasis haematobium had been treated with praziquantel of different dosages by the Chinese Medical Team in Mali. The patients were all confirmed by egg detection in urine examination and were divided into three groups. Group A, 1,187 cases, were each given a single dose of 40 mg/kg of praziquantel. In Group B, 321 cases, were each treated with 10 mg/kg of praziquantel, t.i.d. for two days. In Group C, 119 cases, were each administered with one dose of 50 mg/kg of praziquantel. Three months after treatment, the urine egg negative conversion rates were 88.5%, 96.6% and 89.1%, respectively. It was showed that Group B gave the best efficacy and produced fewer side effects, suggesting that the regime of praziquantel treatment given in divided doses is better than that given at a single dose.

Schistosomiasis chemotherapy.

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After malaria, schistosomiasis (or bilharzia) is the second most prevalent disease in Africa, and is occurring in over 70 countries in tropical and subtropical regions. It is estimated that 600 million people are at risk of infection, 200 million people are infected, and at least 200,000 deaths per year are associated with the disease. All schistosome species are transmitted through contact with fresh water that is infested with free-swimming forms of the parasite, which is known as cercariae and produced by snails. When located in the blood vessels of the host, larval and adult schistosomes digest red cells to acquire amino acids for growth and development. Vaccine candidates have been unsuccessful up to now. Against such devastating parasitic disease, the antischistosomal arsenal is currently limited to a single drug, praziquantel, which has been used for more than 35 years. Because the question of the reduction

of the activity of praziquantel was raised recently, it is thus urgent to create new and safe antischistosomal drugs that should be combined with praziquantel to develop efficient bitherapies.

[Urinary schistosomiasis: report of a case].

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A 31-year-old Japanese man visited our department with macroscopic hematuria. He had stayed in Asia, Middle East and Africa during the past several years. He was diagnosed with urinary schistosomiasis because examination of urine and a biopsy specimen of the bladder mucosa revealed *Schistosoma haematobium* ova. Cystoscopy revealed granulomatous lesions and polypoid patches at the right ureteral orifice, posterior wall and dome of the bladder. Radiological examination showed no obstructive uropathy. Histologically, moderate to severe urothelial dysplasia was found but without evidence of malignancy. The disease has been under good control following treatment with praziquantel. Today international exchange between Japan and other countries is common. Therefore, physicians should consider the possibility of schistosomiasis in patients who have been to endemic areas.

Efficacy and side-effects of two praziquantel treatments against *Schistosoma haematobium* infection, among schoolchildren from Côte d'Ivoire.

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Praziquantel is the current drug of choice for the control of schistosome-attributable morbidity and is likely to remain so for several years. However, as there is concern that schistosomes might develop resistance to the drug, the monitoring of praziquantel's efficacy in different epidemiological settings is recommended. Here, the results of an innovative study of the drug's effectiveness, in an area in Côte d'Ivoire that is highly endemic for *Schistosoma haematobium*, are reported. Each of the subjects (354 schoolchildren aged 5-15 years) was given two oral doses of praziquantel (each of 40 mg/kg) 4 weeks apart. The numbers of schistosome eggs in urine samples collected over several consecutive days prior to the first and after the second treatment were then determined. High cure and egg reduction 'rates', of 93.0% and 96.6%, respectively, were found. Although mild and transient side-effects were frequently observed after the first treatments, no severe systemic complaints were recorded. When the 20 children who remained egg-positive after the second dose were each given a third dose of praziquantel, 16 (80%) of them responded and became egg-negative. Unfortunately, the schistosome strains infecting the remaining four children could not be investigated in detailed laboratory studies, because of the failure of eggs to hatch. There were therefore no unambiguous signs of resistance to praziquantel in this epidemiological setting. The benefits and disadvantages, compared with single-dose treatments, of administering praziquantel twice within a few weeks are discussed. It is anticipated that this approach might prove efficacious in areas of high infection intensity. The integration of such pharmaceutical measures with other readily available control tools is likely to mitigate the current, intolerable burden of schistosomiasis.

Dose-finding study for praziquantel therapy of *Schistosoma haematobium* in Coast

Province, Kenya.

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To assess the efficacy of low dose praziquantel regimens in comparison with standard 40 mg/kg dosing in the treatment of urinary schistosomiasis, a random allocation dose-finding trial was performed in children and adults from a *Schistosoma haematobium* endemic region in Coast Province, Kenya. Following an initial screening, 280 individuals with greater than or equal to 50 eggs/10 ml urine were randomly assigned to receive either 10, 20, 30, or 40 mg/kg of the drug in a single oral dose. Two to three months later, cure rates of 26%, 68%, 78%, and 84% were found for the 10, 20, 30, and 40 mg/kg doses, respectively. The results of 10 mg/kg oral dosing were significantly worse than for all other doses in terms of cure rate and of post-treatment prevalence of morbidity. The 40 mg/kg dosing resulted in a significantly higher cure rate than the 20 mg/kg doses; nevertheless, there was no significant difference between 20 mg/kg and 40 mg/kg doses in terms of mean post-treatment intensity of infection or post-treatment prevalence of hematuria or proteinuria. For large-scale control programs, oral 20 mg/kg praziquantel therapy for urinary schistosomiasis may prove as effective as the standard oral 40 mg/kg dosing for control of infection-associated morbidity and reduction of parasite transmission.
